

Regulating health data: an economic analysis of data protection law

Abstract

The rights to access, share and process health data are increasingly recognised as valuable resources needed to be allocated wisely. Health data recorded, kept, and used by healthcare practitioners (HCP) may cause externalities to consumers, which are internalised with the consumer's exclusive, transferrable quasi-property right to their health data. How health data can be used is restricted by the implicit or explicit contract between the two parties, where public trust plays a vital role in facilitating the transaction. While information campaigns and education fail to promote health data usage, transactional cost economics aims to calculate the cost and benefits of each resource allocation method, enabling policy insights into public trust building for health data. Based on the economic analysis, the government should facilitate public trust by minimising the risk and uncertainty for health data transactions with flexible yet ethical contractual arrangement options and carefully considered and implemented cyber insurance plans. Such arrangements can minimise the risk from externalities and compensate those who suffer more efficiently than litigation, thus building public trust and allowing the market to achieve social efficiency through competition. Importantly, health data regulation should not be solely politically driven and must be considered carefully to ensure that it brings more benefits than its cost.

Introduction

Accessing, sharing and processing health data is essential to healthcare and medical research, and inaccurate or incomplete data can result in misjudgement that undermine individual and population health outcomes, including disease outbreaks and fatalities. Public trust in healthcare plays a significant role in promoting the exchange of health data, which can ultimately benefit society by enabling medical research. However, low levels of public trust, primarily driven by privacy concerns, can lead to the abandonment of data aggregation plans, as evidenced by the termination of the care.data program by the NHS in 2016.^{1,2} The implementation of the General Data Protection Regulation (GDPR) in 2018 has granted consumers more rights to protect their data privacy and imposed stricter regulations on various market entities, leading to significant compliance costs. Recent research by Janßen et al. (2022) indicates that the GDPR has resulted in reduced consumer surplus and the aggregate use of innovative applications, indicating that the costs of data protection cannot be justified economically.³ Moreover, recent studies suggest that the GDPR has not increased public trust in data collection as anticipated,^{4,5} further questioning the current approach to health data regulation.⁶

As Arrow (1963)⁷ states, *"It is clear from everyday observation that the behaviour expected of sellers of medical care is different from that of business."* Regulation on accessing, sharing and processing healthcare data is widely perceived as a legal and political issue rather than an economic one, as there is an expectation of the absence of profit-seeking behaviour in healthcare. Yet, this expectation is unrealistic as various HCPs are for profit, despite their controversial role,⁸ and insufficient income will also drive the profit-seeking behaviour of doctors in public hospitals.⁹ Thus, healthcare can and should be analysed from an economic perspective. As the healthcare system collects various patient-level data from medical records, genetic sequencing, physical tests, etc., the extensive interest in the data has led to privacy concerns.¹⁰ As the emphasis on data privacy has been diffused to more jurisdictions regardless of the status of human rights protection, there exists a universality beyond politics.^{11,12} Thus, we believe there is a need to understand the regulatory issue for data privacy from an economic perspective, to provide a new insight into health data regulation.

Fundamentally, deciding who can use health data and how is an issue of entitlement. As Calabresi and Melamed (1972)¹³ states, *"The state not only has to decide whom to entitle, but it must also simultaneously make a series of equally difficult second order decisions. These decisions go to the manner in which entitlements are protected and to whether an individual is allowed to sell or trade the entitlement."* Thus, we will first elucidate legal definitions of the rights to health data, which will enable further economic analysis. Eventually, we show that there shouldn't be a one-size-fit-all solutions for health data regulation, while diverse contractual arrangements should be allowed to maximise social efficiency, in which the government should carefully consider costs and benefits.

Data privacy: from exclusiveness to transferability

In the world of law and economics, whether one has the right to use something often depends on who owns it, while the regulators need to clarify and ensure the right.¹³ Suppose you are an artist who creates artwork by painting random numbers, which entitles to look at and modify any number in your artwork and create the artwork as you wish, as you have a property right to it, even if people dislike the way you create it. However, how you might be allowed to do so will be up to the purchaser if the artwork has been pre-ordered and purchased. Any unauthorised access, modification and processing of the artwork will be a violation of property law, for which you will be held responsible, even though you cause no harm to the artwork and its owner. Regulators will thus define what the owner can do to protect their rights either by themselves or through public institutions, such as litigation. Property right could protect intangible items in its form but not the derivatives. Suppose that your work is displayed at a gallery visited by a mathematician, who find a pattern in the artwork which you don't even realise. Inspired by the pattern, the mathematician reveals an important theorem, which allows him to win numerous awards and earn a fortune. Should the artist have a share in his success? People may not know the artist until the mathematician mention the artist. However, if the mathematician creates a series of recognisably similar artworks, they will be held responsible for violating the intellectual property of you.

Thus, the question of to what extent how accessing, sharing and processing of health data should be regulated depends on who has what right. Unfortunately, there is no clear answer to this due to the dubious ownership of health data.¹⁴ One might question whether we really have the exclusive right to the health data as most health data are produced, recorded and kept by HCP, during which consumers may not fully understand or beware of how the data are produced and meant for. We grant the right for HCP to study our body, from which HCP derive their own knowledge and understanding, which enable them to provide the service we pay for. In this sense, the consumer's exclusive right to the data, or the property right, is not self-evident,¹⁴ and also questionable in the common law. In *Moore v. Regents of the University of California*,¹⁵ the cells reproduced from patient tissues did not yield any property rights for Moore, as the cells are "*no more unique to Moore than the number of vertebrae in the spine or the chemical formula of haemoglobin*" while expanding the right may impede medical research. Similarly, patient-level data may not be unique to the person, as each individual may produce tonnes of data, e.g. heartbeats, daily, representing fixed, existing patterns. The patterns have broad medical implications, while restricting access to the data may delay diagnosis and treatment, undermining individual and population health outcomes. Thus, whether we own our healthcare data is a highly contested issue.

Modern-day data privacy rights are actually quasi-property rights that avoid the questionable ownership, with the emphasis on exclusiveness and transferability. Despite the ownership dispute, privacy rights are a more common notion regarding health data protection, of which the origin is closely associated with property rights. As quoted in *Semayne's Case* (1604),¹⁶ "*The house of everyone is to him as his castle and fortress, as well for his defence against injury and violence, as for his repose.*" The exclusiveness of private, peaceful space is an early concept of

privacy rights.¹⁷ Warren and Brandeis (1890)¹⁸ proposed separate privacy rights from property rights. The term "reasonable expectation of privacy" was coined in *Katz v. United States* (1967),¹⁹ which established the right to preserve his privacy in a public place, e.g. a telephone booth in the case, if there was a socially recognised expectation of privacy of the individual. Throughout history, our claim on privacy has been restricting rather than controlling. We do not own the telephone booth or the conversation there, but our privacy would be breached if someone deliberately eavesdropped the conversation. Similarly, a health insurance company was not entitled to reveal and transfer their data about a particular person to another party, as the data are their privacy, emphasising the data's exclusiveness. What changes this is the Health Insurance Portability and Accountability Act (HIPPA) of 1996 provides similar rights and responsibilities to HCP and consumers, yet it also emphasises the transferability of health data, with the aim of alleviating job locks due to the lack of data portability between health insurance providers by allowing consumers to transfer the data to another service provider.

As we review the emergence of privacy rights from property rights, we note that privacy protection stems from an expected obligation of confidentiality in implicit or explicit contracts, upon which people form principal-agency relationships in government or healthcare services. These contracts set out the term and conditions for transactions, including health data exchanges. This explanation is more evident under English law, where privacy is protected under breach of confidence. In *Prince Albert v Strange* (1849), the court found that unauthorised transfer, copy, or publication "a breach of trust, confidence, or contract, would of itself entitle the plaintiff to an injunction."

Therefore, the economic nature of privacy and confidentiality is a contractual obligation derived from the transactions. As Cheung (1968)²² notes, different contract arrangements "do not imply different efficiencies of resource allocation as long as property rights are exclusive and transferable." Thus, transactions do not necessarily involve explicit property rights and solid ownership claims as long as the rights are both exclusive and transferable. The transferability of personal health data under HIPPA has yielded a positive impact on job locks,²³ while protecting the data alone is insufficient to achieve transactions, as in the pre-HIPPA age. A clearly defined property right means more tradable values. Thus, a mercantilist government will prevent the export of such values and is likely to demand more of such rights as tradable resources domestically, partly explaining the political motivation for defining quasi-property rights and giving rise to the issue of data sovereignty.²⁴ The negative impact on foreign business of GDPR exemplifies its role as a technical barrier to trade,²⁵⁻²⁷ which also reflects the mercantilist tendency underlying data right claims.

Rationalising health data regulation

Economists often hold different views from politicians in terms of public policy due to different agendas and priorities to achieve. While politicians look into the purposes and aims of each health data usage, to ensure its political correctness and moral standards, economists often aim to maximise social efficiency through a well-functioning competitive market. Such a market is often considered a complex, dynamic system, in which rational agents pursue their own purposes and aims, good or bad. Given perfect competition, no externalities, complete

information, and no transaction costs, a competitive market, as an 'invisible hand', will eventually result in a Pareto-efficient allocation of resources, where it is possible to expand the overall gains without causing harm to others. What the government should do is to let the market work freely by respecting individual rights.

As we find that the limitedly defined property rights to privacy are sufficient to enable health data transactions, we should recognise that the appropriate use of health data is an issue of resource allocation, where individuals and HCP have competing interests. The individual-level health data used by HCP, while providing biomedical insights and advancing social welfare,^{28,29} may also have externalities on the individual and wider population,³⁰ such as social discrimination.³¹ While patients claim data rights to protect themselves from externalities, their avoidance of data sharing also affects the health system, making it less efficient and less accurate, which undermines public trust in healthcare and reduces revenue from healthcare services. Thus, the market fails to approach the maximum interests due to the existence of externalities. The extent of data collection, management and processing is restricted by the externalities.

However, as Arrow (1963)⁷ and Pauly (1978)³² note, the healthcare market is characterised by imperfect competition due to barriers to market entry, exit and technology investment, externalities in public health issues, and information asymmetry between patients and doctors. Due to the abovementioned limitations refuting the assumptions of the foundational theorems of welfare economics, the market cannot reach its optimum. Therefore, welfare economists typically support regulation through redistribution of initial resource allocation needs to be in place for best social efficiency. One of the ways to do so is to empower consumers with more rights and providing them with more information, which will enable rational agents to negotiate at their best interests. If healthcare resources are perceived to be distributed fairly and efficiently, this may lead to greater public trust in the healthcare system. In this sense, public trust is based on receiving and understanding the information, as they believe that given the right information, people will believe that they are doing the right thing with their data rights.

Welfare economists' focus on social efficiency is often impeded by their inability to overcome the human nature of bounded rationality and opportunism, which makes individuals suspicious about the desired social efficiency. Coase (1960)³³ argues that each transaction in the market comes with a cost. In every transaction, rational agents will ensure that the cost should not exceed the benefit of the transaction. Only when the transactional cost is low can the market approach maximum efficiency. Assuming both health data and health services are provided by the same agent, each agent will provide quality care to themselves and keep their data safe, which itself is Pareto-efficient. When externalities exist, there will be a need for a transaction between data and service. Whether there is a data exchange depends on whether the two agents can find each other, how they bargain, what the cost is to keep the data or receive medicine, etc. These costs are summarised as search and information costs, bargaining and decision costs and policing and enforcement costs.

According to the Coase theorem,³⁴ in a competitive market without these externalities, rational agents will achieve a market equilibrium of Pareto efficiency, regardless of the initial resource distribution. Legal arrangements should

be made to lower the costs.³³ Thus, from a transactional cost economics perspective, current data protection laws, such as GDPR, does not provide a framework to quantify the externalities. Thus, it is hard to know whether the strict liability reduces transactional costs and maximises social efficiency. As there is no safe way to completely prevent data breach, for which data dealers are held legally responsible, the strict liability of the privacy breach will lead to repeated compensation claims, which will impede future transactions.

Advancing public trust: a risk-based approach

In a welfare state, the sustainability of the social welfare depends on the willingness of agents to pay the tax that support social welfare, which is affected by the trust in other people making the same contribution.³⁵ Thus, welfare economists uses education, information campaign to facilitate individual decision making. In the case of care.data, we can observe on Twitter that extensive information campaign does not always work.³⁶ To the government, they may also use a mandate. Yet, enforcing a stringent privacy law is similar to the lighthouse, a public good is expensive to maintain and challenging to charge the users. The cost will eventually be paid through universal taxation or fining non-compliant companies. Without proper restrictions, there might be an urge to profit from the fines for governments.

In contrast, transactional cost economics will try to estimate the reciprocal impact of an externality in each transaction. Healthcare usage and health data sharing are fundamentally a matter of transaction, where healthcare- and privacy-related behaviours are characterised by its high uncertainty.^{32,37} Medical researchers use tonnes of data, whilst very little of it can be translated into new therapeutics that benefit or harms the patients. To deal with the externality, the patent law is introduced to prevent free riders, which has been discussed by Ramello (2010).³⁸ Similarly, when an individual provides their data, they may hardly realise whether and how their data might be breached and what they are used for, as in the Facebook-Cambridge Analytica scandal.³⁹ Thus, it is very hard to estimate the gains and losses from health data sharing when people make the decision, due to the risk.

As Das and Teng (2004) notes, "subjective trust and perceived risk are in fact mirror images of each other."⁴⁰ Thus, the public trust in data exchanges is closely associated with perceived risk. Public trust plays a vital role in lowering transactional costs, which reflects the risk from opportunism.⁴¹ Enforcing a stringent data protection law within a well-functioning health system will increase public trust in health data, as it will increase the transparency of personal data usage. However, a transparent but ill-functioning system will expose opportunism within system, which adds to privacy concerns and data protection, leading to the tragedy of the anticommons,⁴² where overclaimed rights impede data sharing and innovations. The health system in a democracy is typically highly transparent due to the free expression of the media, health providers, and consumers, which exposes potential risks from data sharing, as in the case of care.data. Thus, policy makers should also include minimising risk perception as their priorities.

Provided the risk and uncertainty, the common law enforces strict liability to protect one's right to health and privacy,^{43,44} to minimise the risk. For example, the doctors are required to receive strict training, while unlicensed medicine is a crime even when the doctor harms no one. Similarly, the consumers are allowed to sue the data collectors even when the data breach causes no actual damage to any individual involved. Thus, HCP typically need to pay a huge compliance cost, for their training, licensing, etc., before they are eligible to profit from healthcare, which also significantly reduces the available providers and leads to imperfect competition in the healthcare market. Suppose we take the same approach to regulating health data, we might expect the same result due to the huge compliance cost ruling out small competitors.

Thus, the government should allow more flexible regulatory frameworks for data transactions rather than one unified framework, which enables consumers and service providers to quantify the differences in risk by themselves. The right to access, share, and process healthcare data may be treated differently on personal needs, given the diverse usage of health data. For example, the government could require different data protection levels for large, medium and small businesses, restricting the data type and size exposed to each level. Health data collectors and analysers should apply different frameworks to their core and non-core services, thus enabling the user to use their services with minimal data required. The government may facilitate the transaction by providing standard contract templates for each liability level, with clearly defined legal requirements.

For both health data users and providers, there will be a huge cost for managing the risk from data breach. A common way to deal with uncertainty is insuring the negative event. For example, GDPR-specific insurance has now provided in 8 EU jurisdictions for HCP,⁴⁵ while more cyber insurance plans exist.⁴⁶ Thus, the government can facilitate well-designed insurance plans that promote risk management practice with moral hazards taken into consideration. A recent example is the UK government's Cyber Essential scheme, which combines cyber insurance and technical support and accreditation for cybersecurity.⁴⁷ This approach will be more efficient than litigation for businesses, governments and each individual, but more could be done to support small businesses and individuals with more flexible, efficient contractual arrangements.

Conclusion

As Schwab (1989) says, "Private parties do not eliminate externalities when the transaction costs of doing so exceed the benefits. Likewise, however, the costs of government intervention may exceed the benefits." Health data protection presents a complex challenge for both policymakers and economists. To what extent health data should be protected is a matter of various entitlements for health data. Despite the success in raising awareness, the current data protection policy, such as GDPR, may not be fair and efficient economically. A risk-based approach should be used to advance public trust, rather than the sole emphasis on education and information campaigns. A more flexible regulatory approach that enables consumers and service providers to quantify differences in risk and transact accordingly. Insurance and standard contract templates may also provide more efficient solutions than

litigation. Ultimately, policymakers must carefully consider the costs and benefits of government intervention in protecting health data, considering the various economic theories and approaches.

References

1. Staa TP van, Goldacre B, Buchan I, Smeeth L. Big health data: the need to earn public trust. *BMJ*. 2016;354:i3636. doi:10.1136/bmj.i3636
2. Godlee F. What can we salvage from care.data? *BMJ*. 2016;354:i3907. doi:10.1136/bmj.i3907
3. Janßen R, Kesler R, Kummer ME, Waldfogel J. GDPR and the Lost Generation of Innovative Apps. Published online May 2022. doi:10.3386/w30028
4. Bauer PC, Gerdon F, Keusch F, Kreuter F, Vannette D. Did the GDPR increase trust in data collectors? Evidence from observational and experimental data. *Inf Commun Soc*. 2022;25(14):2101-2121. doi:10.1080/1369118X.2021.1927138
5. Strycharz J, Ausloos J, Helberger N. Data Protection or Data Frustration? Individual Perceptions and Attitudes towards the GDPR. *Eur Data Prot Law Rev EDPL*. 2020;6(3):407-421. doi:10.21552/edpl/2020/3/10
6. Marelli L, Lievevrouw E, Van Hoyweghen I. Fit for purpose? The GDPR and the governance of European digital health. *Policy Stud*. 2020;41(5):447-467. doi:10.1080/01442872.2020.1724929
7. Arrow KJ. Uncertainty and the Welfare Economics of Medical Care. *Am Econ Rev*. 1963;53(5):941-973. Accessed April 27, 2023. <https://www.jstor.org/stable/1812044>
8. Street A. For-profit health care might be damaging population health. *Lancet Public Health*. 2022;7(7):e576-e577. doi:10.1016/S2468-2667(22)00142-6
9. Tian Y, Hua LJ, Chao WM. Chinese doctors' salaries. *The Lancet*. 2008;371(9624):1576-1577. doi:10.1016/S0140-6736(08)60684-4
10. Lopes P, Silva LB, Oliveira JL. Challenges and Opportunities for Exploring Patient-Level Data. *BioMed Res Int*. 2015;2015:e150435. doi:10.1155/2015/150435
11. Wang Han S, Munir AB. Information Security Technology - Personal Information Security Specification: China's Version of the GDPR. *Eur Data Prot Law Rev EDPL*. 2018;4:535. <https://heinonline.org/HOL/Page?handle=hein.journals/edpl4&id=567&div=&collection=>
12. Barrett C. Are the EU GDPR and the California CCPA becoming the de facto global standards for data privacy and protection? *Scitech Lawyer*. 2019;15(3):24-29.
13. Calabresi G, Melamed A. Property Rules, Liability Rules, and Inalienability: One View of the Cathedral. *Fac Scholarsh Ser*. Published online January 1, 1972. Accessed April 27, 2023. <https://openyls.law.yale.edu/handle/20.500.13051/1262>
14. Mirchev M, Mircheva I, Kerekovska A. The Academic Viewpoint on Patient Data Ownership in the Context of Big Data: Scoping Review. *J Med Internet Res*. 2020;22(8):e22214. doi:10.2196/22214
15. *Moore v. Regents of Univ. of Cal.* - 51 Cal. 3d 120, 271 Cal. Rptr. 146, 793 P.2d 479. (Supreme Court of California 1990).
16. *Semayne's Case*. Co. Rep. 5, 91 (Court of Queen's Bench 1604). Accessed April 28, 2023. <https://uk.westlaw.com/Document/I0582BB10E57C11DAB242AFE6182DD7E/View/FullText.html>

17. Chlopecki M. The Property Rights Origins of Privacy Rights. Foundation for Economic Education. Published August 1, 1992. Accessed April 27, 2023. <https://fee.org/articles/the-property-rights-origins-of-privacy-rights/>
18. Warren SD, Brandeis LD. The Right to Privacy. *Harv Law Rev.* 1890;4(5):193-220. doi:10.2307/1321160
19. *Katz v. United States*. (supreme.court).
20. O'Mara M. The End of Privacy Began in the 1960s. *The New York Times*. <https://www.nytimes.com/2018/12/05/opinion/google-facebook-privacy.html>. Published December 5, 2018. Accessed May 1, 2023.
21. Packard V. *The Naked Society*. D. McKay Company; 1964.
22. Cheung SNS. Private Property Rights and Sharecropping. *J Polit Econ.* 1968;76(6):1107-1122. doi:10.1086/259477
23. Sanz-De-Galdeano A. Job-Lock and Public Policy: Clinton's Second Mandate. *Ind Labor Relat Rev.* 2006;59(3):430-437. doi:10.1177/001979390605900305
24. Chin YC, Li K. A Comparative analysis of Cyber Sovereignty Policies in China and the EU. Published online August 6, 2021. doi:10.2139/ssrn.3900752
25. Kuchler H. US small businesses drop EU customers over new data rule. *Financial Times*. <https://www.ft.com/content/3f079b6c-5ec8-11e8-9334-2218e7146b04>. Published May 24, 2018. Accessed May 1, 2023.
26. Cory N. *How the EU Is Using Technology Standards as a Protectionist Tool In Its Quest for Cybersovereignty*.; 2022. Accessed May 1, 2023. <https://itif.org/publications/2022/09/19/how-the-eu-is-using-technology-standards-as-a-protectionist-tool/>
27. Meddin E. The Cost of Ensuring Privacy: How the General Data Protection Regulation Acts as a Barrier to Trade in Violation of Articles XVI and XVII of the General Agreement on Trade in Services Comments. *Am Univ Int Law Rev.* 2019;35(4):997-1036. Accessed May 1, 2023. <https://heinonline.org/HOL/P?h=hein.journals/amuilr35&i=1037>
28. Institute of Medicine. Healthcare Data: Public Good or Private Property? In: Goolsby WA, Olsen L, McGinnis JM, eds. *Clinical Data as the Basic Staple of Health Learning: Creating and Protecting a Public Good: Workshop Summary*. National Academies Press; 2010. Accessed April 27, 2023. <https://www.ncbi.nlm.nih.gov/books/NBK54304/>
29. Larson EB. Building Trust in the Power of "Big Data" Research to Serve the Public Good. *JAMA.* 2013;309(23):2443-2444. doi:10.1001/jama.2013.5914
30. Ferguson B, Keen J. Transaction costs, externalities and information technology in health care. *Health Econ.* 1996;5(1):25-36. doi:10.1002/(SICI)1099-1050(199601)5:1<25::AID-HEC181>3.0.CO;2-F
31. Favaretto M, De Clercq E, Elger BS. Big Data and discrimination: perils, promises and solutions. A systematic review. *J Big Data.* 2019;6(1):12. doi:10.1186/s40537-019-0177-4
32. Pauly MV. Is medical care different. In: *Competition in the Health Care Sector: Past, Present, and Future: Proceedings of a Conference Sponsored by the Bureau of Economics, Federal Trade Commission, March 1978.* ; 1978.
33. Coase RH. The Problem of Social Cost. *J Law Econ.* 1960;3:1-44. doi:10.1086/466560

34. Felder J. Coase Theorems 1-2-3. *Am Econ*. 2001;45(1):54-61. doi:10.1177/056943450104500105
35. Algan Y, Cahuc P, Sangnier M. Trust and the Welfare State: The Twin Peaks Curve. *Econ J*. 2016;126(593):861-883. doi:10.1111/eoj.12278
36. Hays R, Daker-White G. The care.data consensus? A qualitative analysis of opinions expressed on Twitter. *BMC Public Health*. 2015;15(1):838. doi:10.1186/s12889-015-2180-9
37. Acquisti A, Loewenstein G, Brandimarte L. Privacy and human behavior in the age of information. *Science*. Published online 2015. doi:10.1126/science.aaa1465
38. Ramello GB. Property rights and externalities: the uneasy case of knowledge. *Eur J Law Econ*. 2011;31(1):123-141. doi:10.1007/s10657-010-9198-3
39. Knowledge at Wharton Staff. Why the Cambridge Analytica Scandal Is a Watershed Moment for Social Media. Knowledge at Wharton. Published March 22, 2018. Accessed April 30, 2023. <https://knowledge.wharton.upenn.edu/article/fallout-cambridge-analytica/>
40. Das TK, Teng BS. The Risk-Based View of Trust: A Conceptual Framework. *J Bus Psychol*. 2004;19(1):85-116. doi:10.1023/B:JOB.0000040274.23551.1b
41. Williamson OE. Calculativeness, Trust, and Economic Organization. *J Law Econ*. 1993;36(1, Part 2):453-486. doi:10.1086/467284
42. Ying Q. Anticommons, Tragedy of the. In: Marciano A, Ramello GB, eds. *Encyclopedia of Law and Economics*. Springer; 2019:65-71. doi:10.1007/978-1-4614-7753-2_90
43. Cox AHL. The Medical Profession and Strict Liability for Defective Products--A Limited Extension. *Hastings Law J*. 1965;17:359. <https://heinonline.org/HOL/Page?handle=hein.journals/hastlj17&id=391&div=&collection=>
44. Sundholm B. Strict Liability for Genetic Privacy Violations in the Age of Big Data. *Univ Memphis Law Rev*. 2018;49:759. <https://heinonline.org/HOL/Page?handle=hein.journals/umem49&id=777&div=&collection=>
45. Leemans V, Molony D. Are GDPR Fines Insurable? *Risk Manage*. 2018;65(9):28-30. Accessed May 1, 2023. <https://go.gale.com/ps/i.do?p=AONE&sw=w&issn=00355593&v=2.1&it=r&id=GALE%7CA558823709&sid=googleScholar&linkaccess=abs>
46. Talesh SA. Data Breach, Privacy, and Cyber Insurance: How Insurance Companies Act as “Compliance Managers” for Businesses. *Law Soc Inq*. 2018;43(2):417-440. doi:10.1111/lsi.12303
47. Rafferty I. UK government scheme Cyber Essentials insurance is ‘appalling’ for SMEs. Insurance Times. Published January 30, 2023. Accessed May 2, 2023. <https://www.insurancetimes.co.uk/news/uk-government-scheme-cyber-essentials-insurance-is-appalling-for-smes/1443637.article>